SWE2011– Big Data Analytics

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PROFESSOR GRADE 1
SCORE

Big Data Analytics – Course outcome

- **Recognize of various data analytical techniques and approaches for handling big data.**

MODULE 2 :Data Analysis and Approaches

- The process involves collecting, cleaning, and analyzing data using techniques like machine learning and statistics, and it is categorized into four main methods ie descriptive, diagnostic, predictive, and prescriptive.

MODULE 2 :Data Analysis and Approaches

- Evolution of analytic scalability – Convergence – parallel processing systems – analytic data sets – Analytic methods – Analysis approaches – Statistical significance – business approaches – Analytic innovation – Traditional approaches – Iterative

Basic definition

- **Data:**
- **Known facts that can be recorded and have an implicit meaning.**
- **Database:**
- **Organized collection of related data.**
- **Database Management System (DBMS):**
- **A software package to facilitate the creation and maintenance of a computerized database.**
- **Relational Database Management System (RDBMS):**
- **DBMS based on relational model - Relation is group of tuples**
- **Enterprise Data Warehouse (EDW):**
- **Central warehouse of all sources of data**

Basic definition

- **Analytic Architecture:**
- **Traditional Analytic Architecture**
- **Modern In-Database Architecture**

Basic definition

- **Data Preparation:**
- **Manipulation of data into suitable form for analysis**
- **Join:**
- **Combining columns of different data sources**
- **Aggregation:**
- **Combining all data into one**
- **Combining rows of different data source Eg: statistical summary**
- **Derivations:**
- **Creating new columns of data**
- **Calculating ratio**
- **Transformation:**
- **Converting data into useful format**
- **Example : Taking log, converting date of birth to age**

Basic definition

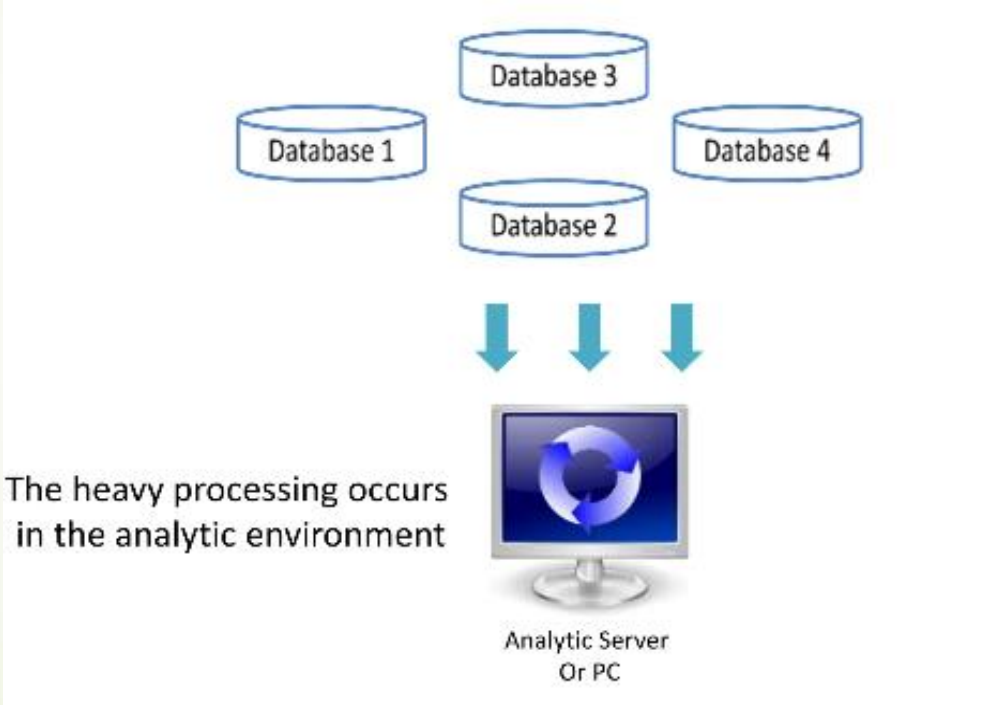
- Ways for in-database data preparation:
- SQL
- User defined functions / Embedded processes
- Analytic tool's engine running on database
- Predictive modeling markup language
- Based on XML

Evolution of analytic scalability

- **Scalability:**
- **The ability of a system to handle increasing amount of work required to perform its task.**
- **The increase in data storage ability has grown in recent years to accommodate the need for big data**
- **Measures of Data Size:**
- **– Kilo, Mega, Giga, Tera, Peta, Exa, Zetta, Yotta**
- **The evolution of analytic scalability moved from traditional, centralized systems to distributed, parallel architectures to handle "Big Data," driven by technologies like Massively Parallel Processing (MPP), Grid Computing, MapReduce, and cloud platforms, enabling processing of massive datasets and complex analytics.**

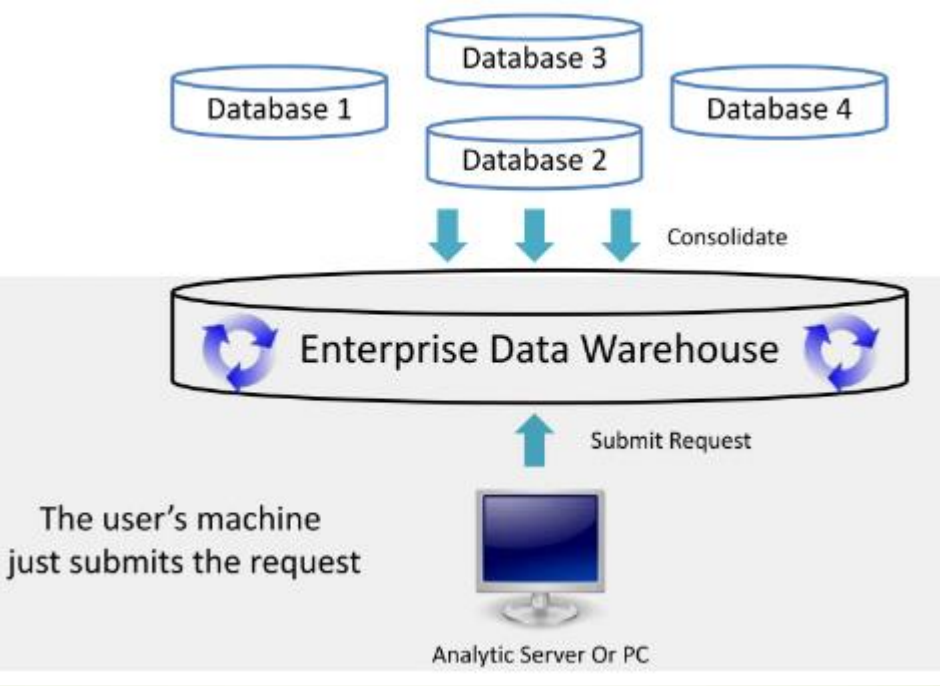
Traditional Analytic Architecture

- **We had to pull all data together into separate analytics environment to do analysis.**



Modern In-Database Architecture

- The processing stays in the database where the data has been consolidated.



MASSIVELY PARALLEL PROCESSING SYSTEMS

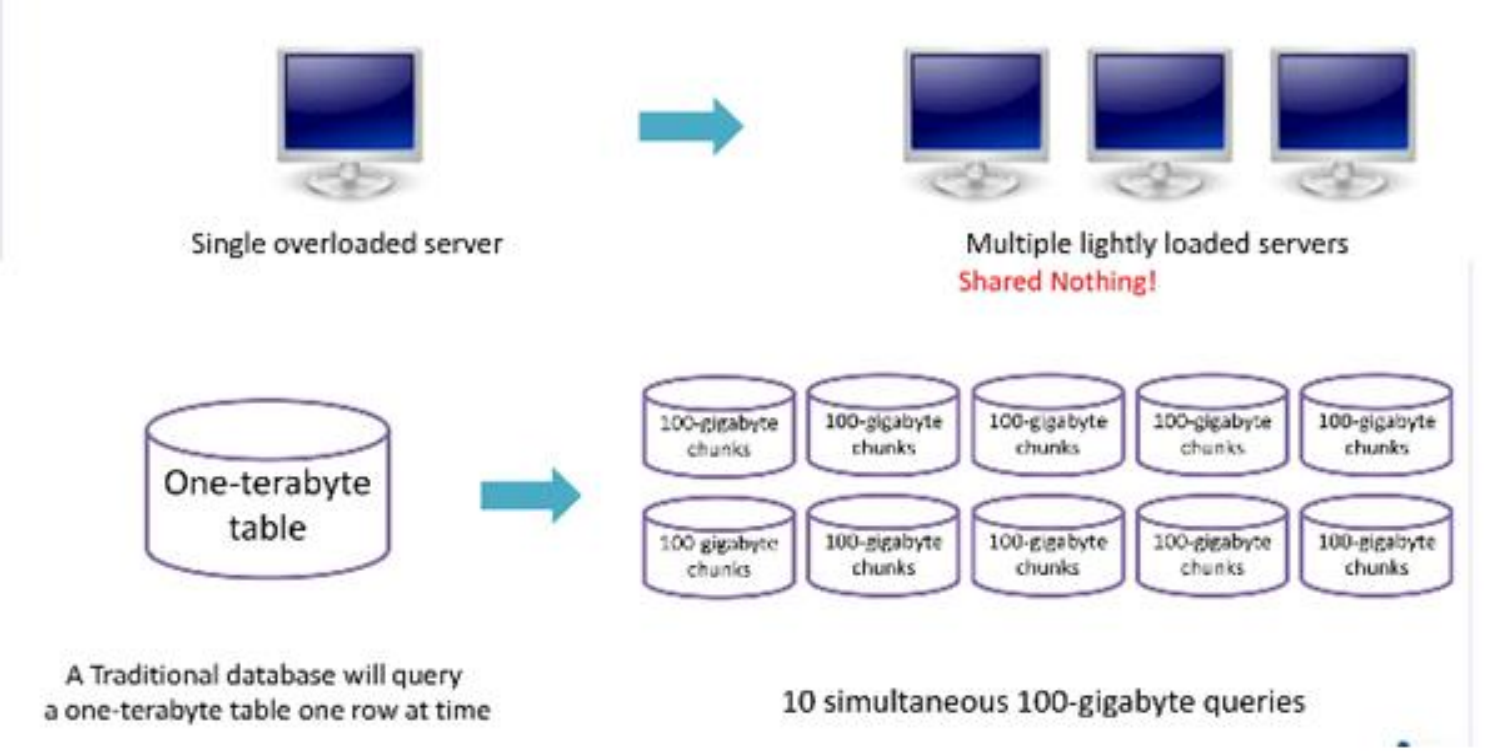
- Has lots of processor
- All these processor works in parallel
- Big data is split into many parts and the processors works in parallel in each part
- Divide and conquer strategy.
- Instead of a single threaded process to work through the data, an MPP system breaks the job into pieces and allows the different sets of CPU and disk to run the process concurrently

MASSIVELY PARALLEL PROCESSING SYSTEMS

- An MPP database spreads data out into independent pieces managed by independent storage and central processing unit (CPU) resources.
- It removes the constraints of having one central server with only a single set CPU and disk to manage it. (Traditional)
- The data in an MPP system gets split across a variety of disks managed by a variety of CPUs spread across a number of servers.

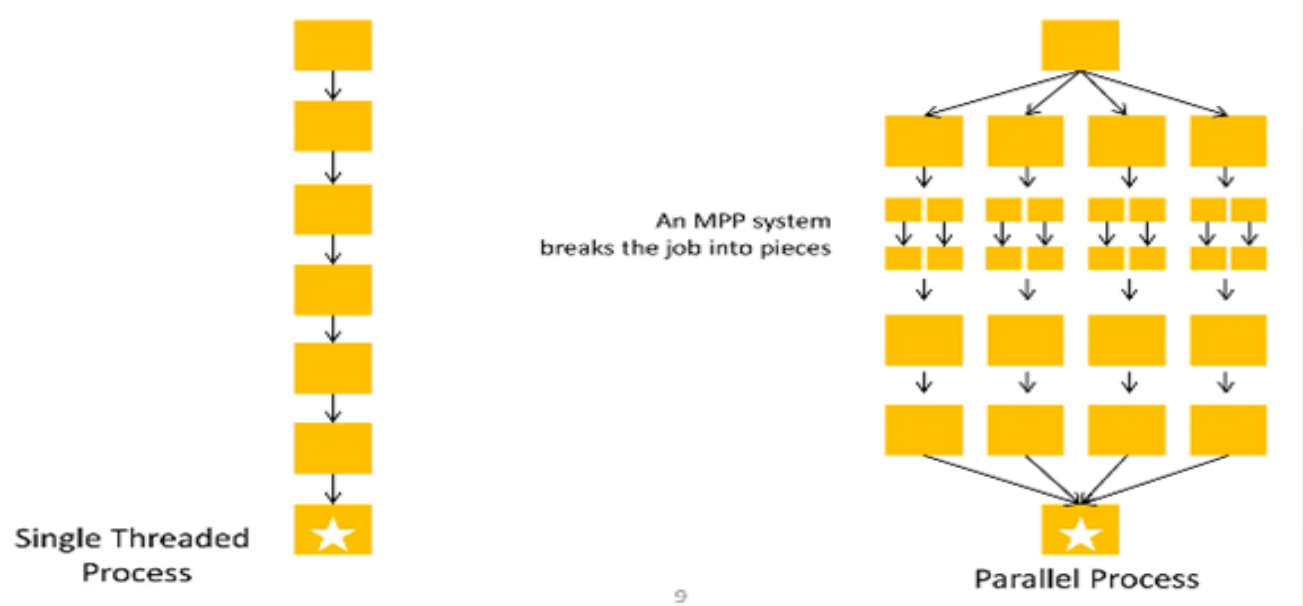
Massively Parallel Processing System

- An MPP database breaks the data into independent chunks with independent disk and CPU.



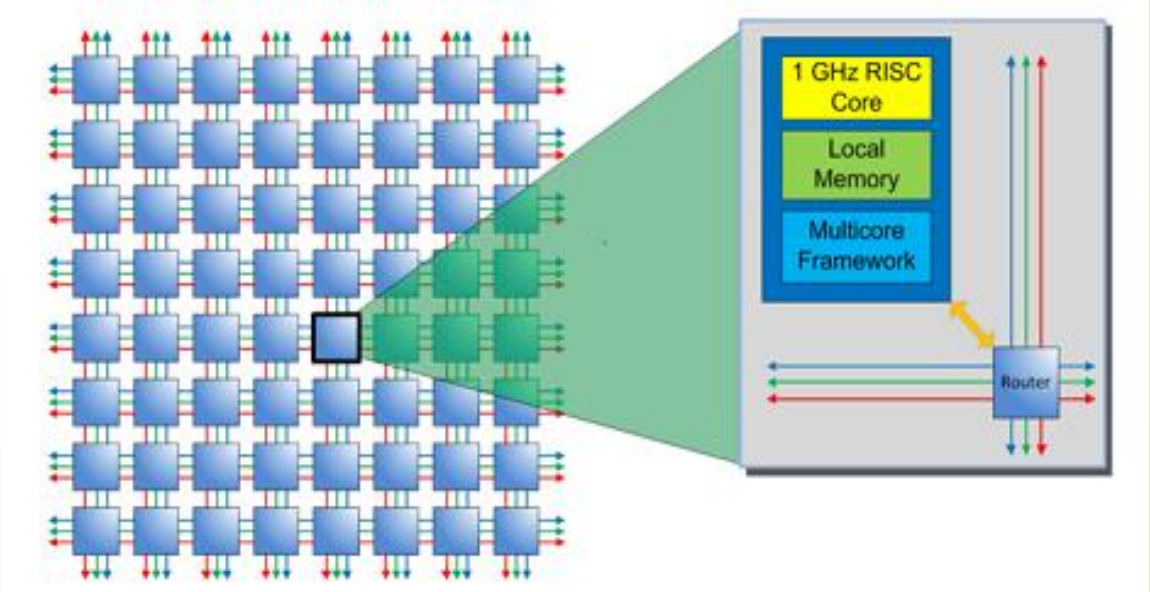
Massively Parallel Processing System

- MPP systems built in redundancy to make recovery easy.
- This systems have resource management tools. Manage the CPU, disk space and Query optimizer.



NEED OF MASSIVELY PARALLEL PROCESSING SYSTEMS

- 1.Computation & storage
- 2. Redundancy
- 3. Resource management tools
- 4. Query optimizers

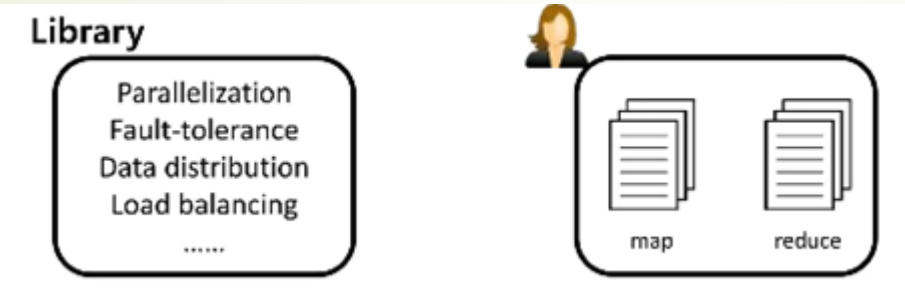


Cloud Computing Environment

- Big data analytics in cloud computing combines the analysis of large, fast, and diverse datasets with the scalable infrastructure of cloud platforms to extract insights efficiently.
- **Data storage:** Cloud platforms provide scalable and cost-effective storage solutions (like data lakes and warehouses) for structured, semi-structured, and unstructured data
- **Data processing**: Powerful, distributed processing tools, such as Apache Spark, run on cloud infrastructure to handle the immense volume and velocity of big data.
- **Analytics**: Cloud-based services and platforms enable advanced analytics techniques like machine learning and data mining to uncover patterns, correlations, and trends from the processed data.
- **Accessibility:** Cloud-based solutions make data and analytics accessible from anywhere with an internet connection, facilitating collaboration and analysis

Map Reduce Framework

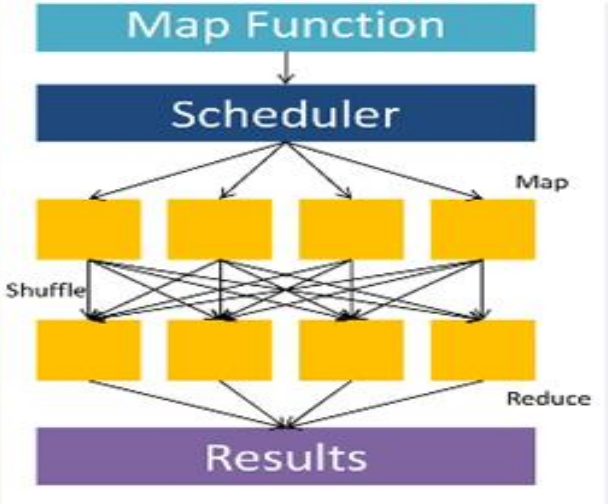
- A parallel programming framework.



- Map function
- Processing a key/value pairs to generate a set of intermediate key/value pairs
- Reduce function
- Merging all intermediate values associated with same intermediate key

Map Reduce Framework

- Distribute a terabyte to each of the 20 nodes using simple file copy process
- Submit two programs (Map, Reduce) to the scheduler
- The map program find the data on disk and executes the logic it contains.
- The result of the map step are then passed to the reduce process to summarize and aggregate the final answers.



ANALYTIC DATA SET

- It also called an analytic or analytical database, is a read-only system that stores historical data on business metrics such as sales performance and inventory levels.
- Business analysts, corporate executives and other workers run queries and reports against an analytic database.
- The information is regularly updated to include recent transaction data from an organization's operational systems.
- An analytic database is specifically designed to support business intelligence (BI) and analytic applications, typically as part of a data warehouse or data mart.
- This differentiates it from an operational, transactional or online transaction processing database, which is used for transaction processing, such as order entry and other business applications.

Sources of analytic databases

- Analytical databases store massive amounts of data that organizations use to gain insight into their business, customers and more.
- The data stored in analytical databases comes from sources such as enterprise resource planning (ERP), customer relationship management (CRM) and other business applications or proprietary data.
- Some analytical databases include a data warehouse, which is designed to be flexible for users who need to create specific reports and queries

Components of analytic databases:

- Two major components of analytical databases are
- Data model
- Query language.
- The data model illustrates how the data is structured within the organization.
- Data models can be
- Relational -- where data is stored in separate tables and joined together when necessary.
- Object-oriented -- where data is stored in fields of objects and represented as objects in software.

Components of analytic databases:

- **Query language.**
- **A query language is a standardized and interpreted programming language for the retrieval of information from a database management system (DBMS) or database.**
- **The information is usually, but not always, in the form of a table or a set of tables.**
- **A query language enables users to retrieve information from a database without knowing the internal structure of the database.**
- **The most well-known query language is Structured Query Language (SQL).**

Types of analytic databases:

- There are five main types of analytic databases
- Columnar databases:
- Massively parallel processing (MPP) databases:
- In-memory databases:
- Online analytical processing (OLAP) databases:
- Data warehouse appliances:

Columnar databases:

- A columnar database stores data in large contiguous blocks of memory called data columns. It is well suited to analytical processing and provide a solution to the problem of data sprawl,
- This differs from a row-oriented database, where data is stored in tables split across columns and rows.
- Columnar databases are designed to store, analyze and retrieve data by column, they can accelerate data loading and improve data analytics performance.
- The columnar architecture of this database also allows for efficient data compression and fast aggregate queries.
- Examples: Amazon Redshift, MariaDB and Apache Kudu.

Massively parallel processing (MPP) databases

- In an MPP database, data is stored on multiple servers rather than on a single server.
- Each server has its own set of data and its own computing power.
- MPP databases typically provide high availability and redundancy by using clusters of computers to serve multiple users.
- They also efficiently handle large amounts of data while providing fast access to that data.
- Examples of MPP databases include Vertica and Teradata.

In-memory databases

- In-memory databases are optimized to store data in random-access memory.
- This contrasts with traditional databases that rely on hard drives to store both data and program instructions. Because data is stored in memory, in-memory databases provide a near-real-time response.
- Storing data in memory offers several potential benefits over traditional databases, including faster query processing, less I/O overhead and the ability to handle larger databases.
- Examples of in-memory databases include SAP HANA, Oracle TimesTen and Aerospike.

Online analytical processing (OLAP) databases

- An OLAP database is an online database that is used to create reports and analyses that are faster and more efficient than traditional databases.
- An OLAP database stores multidimensional cubes of aggregated data for analyzing information based on multiple data attributes.
- The data is stored in a structure designed to facilitate analysis of the data. Because OLAP databases are stored in a multidimensional format, they can handle a high volume of data better than a traditional database.
- Examples of database structures include star schemas and snowflake schemas.
- OLAP databases are used in several industries, including financial services and retail.
- Examples of OLAP databases include Snowflake and IBM Cognos.

Data warehouse appliances

- Data warehouse appliances are integrated hardware and software tools designed to optimize the performance of data warehouses.
- A data warehouse is an information repository large organizations use to centralize, store and index vast amounts of data from disparate sources.
- Data warehouse appliances simplify and expedite data warehouse implementation and management.
- They are commonly used in BI to track and analyze data from many sources, including ERP systems, CRM software and other databases.
- Numerous vendors offer data warehouse appliances, including IBM, Microsoft, Oracle and Teradata.

Data Analysis

- The systematic application of statistical and logical techniques to describe the data scope, modularize the data structure, condense the data representation, illustrate via images, tables, and graphs, and evaluate statistical inclinations, probability data, and derive meaningful conclusions known as Data Analysis.
- Data generation is a continual process; this makes data analysis a continuous, iterative process where the collection and performing data analysis simultaneously.
- Ensuring data integrity is one of the essential components of data analysis.
- examples where data analysis is used: transportation, risk and fraud detection, customer interaction, city planning healthcare, web search, digital advertisement, and more.

Data Analysis Methods

1. Qualitative Analysis:

- This approach mainly answers questions such as 'why,' 'what' or 'how.'
- Each of these questions is addressed via quantitative techniques such as questionnaires, attitude scaling, standard outcomes, and more.
- Such analysis is usually in the form of texts and narratives, which might also include audio and video representations.

2. Quantitative Analysis:

- This analysis is measured in terms of numbers.
- The data here present themselves in terms of measurement scales and extend themselves for more statistical manipulation.

Data Analysis Methods

3. Text analysis:

- Text analysis is a technique to analyze texts to extract machine-readable facts.
- It aims to create structured data out of free and unstructured content.
- The process consists of slicing and dicing heaps of unstructured, heterogeneous files into easy-to-read, manage and interpret data pieces.
- It is also known as text mining, text analytics, and information extraction.
- The ambiguity of human languages is the biggest challenge of text analysis.

Data Analysis Methods

- For example, humans know that "Red Sox Tames Bull" refers to a baseball match. Still, if this text is fed to a computer without background knowledge, it would generate several linguistically valid interpretations.
- Sometimes people who are not interested in baseball might have trouble understanding it too.
- **4. Statistical analysis:**
- Statistics involves data collection, interpretation, and validation.
- Statistical analysis is the technique of performing several statistical operations to quantify the data and apply statistical analysis.
- Quantitative data involves descriptive data like surveys and observational data. It is also called a descriptive analysis.
- It includes various tools to perform statistical data analysis such as SAS (Statistical Analysis System), SPSS (Statistical Package for the Social Sciences), Stat soft, and more.

Data Analysis Methods

- **5. Diagnostic analysis:**
- **The diagnostic analysis is a step further to statistical analysis to provide a more in-depth analysis to answer the questions.**
- **It is also referred to as root cause analysis as it includes processes like data discovery, mining, and drill down and drill through.**
- **6. Predictive analysis:**
- **Predictive analysis uses historical data and feds it into the machine learning model to find critical patterns and trends.**
- **The model is applied to the current data to predict what would happen next.**
- **Many organizations prefer it because of its various advantages like volume and type of data, faster and cheaper computers, easy-to-use software, tighter economic conditions, and a need for competitive differentiation.**

Data Analysis Methods

- **7. Prescriptive Analysis:**
- **Prescriptive analytics suggests various courses of action and outlines the potential implications that could be reached after predictive analysis.**
- **Prescriptive analysis generating automated decisions or recommendations requires specific and unique algorithmic and clear direction from those utilizing the analytical techniques.**

Data Analysis Techniques (Approaches):

- There are different techniques for Data Analysis depending upon the question at hand, the type of data, and the amount of data gathered.
- Each focuses on taking onto the new data, mining insights, and drilling down into the information to transform facts and figures into decision-making parameters.

1. Techniques based on Mathematics and Statistics:

- **Descriptive Analysis:**
- **Descriptive Analysis considers the historical data, Key Performance Indicators and describes the performance based on a chosen benchmark. It takes into account past trends and how they might influence future performance.**
- **Dispersion Analysis:**
- **Dispersion is the area onto which a data set is spread. This technique allows data analysts to determine the variability of the factors under study.**

1. Techniques based on Mathematics and Statistics:

- **Regression Analysis:**
- **This technique works by modeling the relationship between a dependent variable and one or more independent variables. A regression model can be linear, multiple, logistic, ridge, non-linear, life data, and more.**
- **Factor Analysis:**
- **This technique helps to determine if there exists any relationship between a set of variables. This process reveals other factors or variables that describe the patterns in the relationship among the original variables. Factor Analysis leaps forward into useful clustering and classification procedures.**

1. Techniques based on Mathematics and Statistics:

- **Discriminant Analysis:**
- **It is a classification technique in data mining.**
- **It identifies the different points on different groups based on variable measurements.**
- **In simple terms, it identifies what makes two groups different from one another; this helps to identify new items.**
- **Time Series Analysis:**
- **In this kind of analysis, measurements are spanned across time, which gives us a collection of organized data known as time series.**

2. Techniques based on Artificial Intelligence and Machine Learning

- **Artificial Neural Networks:**
- **An Artificial Neural Network is a system that changes its structure based on information that flows through the network. ANN can accept noisy data and are highly accurate. They can be considered highly dependable in business classification and forecasting applications.**
- **Decision Trees:**
- **It is a tree-shaped model representing a classification or regression model. It divides a data set into smaller subsets, simultaneously developing into a related decision tree.**

2. Techniques based on Artificial Intelligence and Machine Learning

- **Evolutionary Programming:**
- **This technique combines the different types of data analysis using evolutionary algorithms. It is a domain-independent technique, which can explore ample search space and manages attribute interaction very efficiently.**
- **Fuzzy Logic:**
- **It is a data analysis technique based on the probability that helps handle the uncertainties in data mining techniques.**

3. Techniques based on Visualization and Graphs

- **Column Chart, Bar Chart:**
- **Both these charts are used to present numerical differences between categories. The column chart takes to the height of the columns to reflect the differences. Axes interchange in the case of the bar chart.**
- **Line Chart:**
- **This chart represents the change of data over a continuous interval of time.**
- **Area Chart:**
- **This concept is based on the line chart. It also fills the area between the polyline and the axis with color, representing better trend information.**

3. Techniques based on Visualization and Graphs

- **Pie Chart:**
- **It is used to represent the proportion of different classifications. It is only suitable for only one series of data. However, it can be made multi-layered to represent the proportion of data in different categories.**
- **Funnel Chart:**
- **This chart represents the proportion of each stage and reflects the size of each module. It helps in comparing rankings.**

3. Techniques based on Visualization and Graphs

- **Word Cloud Chart:**
- **It is a visual representation of text data. It requires a large amount of data, and the degree of discrimination needs to be high for users to perceive the most prominent one. It is not a very accurate analytical technique.**
- **Gantt Chart:**
- **It shows the actual timing and the progress of the activity compared to the requirements.**
- **Radar Chart:**
- **It is used to compare multiple quantized charts. It represents which variables in the data have higher values and which have lower values. A radar chart is used for comparing classification and series along with proportional representation.**

3. Techniques based on Visualization and Graphs

- **Scatter Plot:**
- **It shows the distribution of variables in points over a rectangular coordinate system. The distribution in the data points can reveal the correlation between the variables.**
- **Bubble Chart:**
- **It is a variation of the scatter plot. Here, in addition to the x and y coordinates, the bubble area represents the 3rd value.**
- **Gauge:**
- **It is a kind of materialized chart. Here the scale represents the metric, and the pointer represents the dimension. It is a suitable technique to represent interval comparisons.**

3. Techniques based on Visualization and Graphs

- **Frame Diagram:**
- **It is a visual representation of a hierarchy in an inverted tree structure.**
- **Rectangular Tree Diagram:**
- **This technique is used to represent hierarchical relationships but at the same level. It makes efficient use of space and represents the proportion represented by each rectangular area.**

Map

- **Regional Map:**
- **It uses color to represent value distribution over a map partition.**
- **Point Map:**
- **It represents the geographical distribution of data in points on a geographical background. When the points are the same in size, it becomes meaningless for single data, but if the points are as a bubble, it also represents the size of the data in each region.**

Map

- **Flow Map:**
- **It represents the relationship between an inflow area and an outflow area. It represents a line connecting the geometric centers of gravity of the spatial elements. The use of dynamic flow lines helps reduce visual clutter.**
- **Heat Map:**
- **This represents the weight of each point in a geographic area. The color here represents the density.**
- **Statistical Significance**
- **Statistical significance is a determination by an analyst that the results in the data are not explainable by chance or luck alone.**
- **Statistical hypothesis testing is the method by which the analyst makes this determination.**

Map

- **Statistical Significance**
- **Statistical significance is a determination by an analyst that the results in the data are not explainable by chance or luck alone.**
- **Statistical hypothesis testing is the method by which the analyst makes this determination.**
- **This test provides a p-value, which is the probability of observing results as extreme as those in the data, assuming the results are truly due to chance alone.**
- **A p-value of 5% or lower is often considered to be statistically significant.**

Map

- **Statistical Significance**
- **Statistical significance is a determination that a relationship between two or more variables is caused by something other than chance.**
- **Statistical significance is used to provide evidence concerning the plausibility of the null hypothesis, which hypothesizes that there is nothing more than random chance at work in the data.**
- **Statistical hypothesis testing is used to determine whether the result of a data set is statistically significant.**

Business approaches:

- 1. Traditional
- 2. Big data

Traditional Approach

- 1. Database is based on fixed schema, work efficiently for structured data.
- Need to use different methods for unstructured data
- 2. Analysis done on historical data
- 3. Data is stored in centralized servers
- 4. Sources or origin of data are limited
- 5. Users determine the questions at start. Since structured data queries will be written initially.

Big Data Approach

- 1. Database is based on dynamic schema, work efficiently for all data.
- 2. Analysis done on real time data
- 3. Data is stored in distributed servers
- 4. Data come from different sources(external, internal)
- 5. Users can explore the data and determine the questions – iterative and exploratory approach

Data analytics Approaches

Approach	Definition	Answers the questions
Descriptive	Describes and summarizes data to gain insights into what has happened in the past.	• What happened? • How many?
Diagnostic	Identifies the root cause of an issue or problem	• Why did it happen? • What caused it?
Predictive	Analyzes historical data and makes predictions about future events.	• What is likely to happen? • What if?
Prescriptive	Provides recommendations for actions you should take based on the analysis.	• What should we do? • How can we improve?